

Running to Stand Still: Shadow Education and Human Capital Formation*

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Abstract

Exploiting spatiotemporal variation in the entry of private after-school tutoring centers across Chinese cities from 1993 to 2018, we estimate the causal impact of tutoring access on student outcomes. Elementary school tutoring has no significant effect on middle school enrollment. Middle school tutoring raises high school enrollment, particularly among girls, disadvantaged students, and rural populations, yet does not translate into higher high school graduation or college enrollment rates. High school tutoring, concentrated in more developed areas, increases college enrollment, potentially widening educational disparities. Despite these enrollment gains, tutoring access does not improve graduate school admissions or future earnings—even though families, particularly disadvantaged ones, substantially increase their educational spending in response to center openings. Investigating mechanisms, we show that private tutoring improves cognitive skills but not non-cognitive skills, helping explain its limited long-run returns.

JEL classification: I21, I22, I24, I26, J24

Keywords: Shadow education, human capital formation, cognitive and non-cognitive skills, educational inequality, returns to education

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